

Robotics AI Engineering Intern

Perception, Simulation & Robot Integration

AIMRO | Herndon, Virginia, Hybrid | Part-time | Paid

ABOUT THE WORK

AIMRO builds AIMEX, an AI platform that teaches industrial robots to perform assembly tasks by watching people do them. Our lab is where that gets real: cameras, robots, simulation, and vision models, all working together to turn human demonstrations into robot actions. We're a small team, and that's the best part: nobody is boxed into one lane, and you'll get your hands on far more of the stack than you would anywhere else.

WHAT YOU'LL DO

This is an apprenticeship in becoming a full-stack robotics AI engineer. Over time you'll work across the whole loop:

- **Perception:** train and evaluate vision models on synthetic and real data, and close the sim-to-real gap.
- **Simulation:** build simulation scenes that generate synthetic training data.
- **Robot integration:** bring up cameras, grippers, and robot arms, calibrate them, and get pick-and-place running end to end.
- **Demonstration capture:** run and improve the multi-camera recording rig and the tooling around it.
- **AI-assisted development:** use AI agents as a daily part of the workflow, and help build the knowledge base they run on.
- **Documentation and hardware:** write the design docs, design and 3D-print fixtures and mounts when the lab needs them.

WHO WE'RE LOOKING FOR

Someone who wants to be capable across all of the above and can show it:

- You've built something physical that runs code (a robot, a camera rig, a Raspberry Pi project) and can tell us what broke and how you fixed it.
- Solid Python or C++, and comfortable working from vendor SDKs with thin documentation.
- Comfortable in Linux and Windows, and good at getting hardware to talk.
- Computer vision fundamentals from coursework or projects.
- Already uses AI coding tools, and can explain what they get wrong.

Nice to have: simulation or ROS 2 experience, CAD and 3D printing, Chinese proficiency (much of our vendor ecosystem is Chinese), robotics club or competition experience.

WHERE THIS GOES

We're hiring an intern to grow, not to fill a seat. The path is intern → engineer → owning the lab's technical stack: the person who decides how we capture data, train models, and integrate robots. Want to go deep and become our expert in one of these areas? We'd love that, and we'll help you get there. Want to learn how every piece fits together and eventually run the whole stack? That's the path we're building this role around. Either way, you'll own real work from day one.

LOGISTICS

Flexible working hours, with option to extend or convert. Compensation is based on qualifications and experiences. Hybrid work with a site in Herndon, Virginia. Students in CS, EE, ME, or robotics and recent graduates are welcome.

TO APPLY

Send a resume plus one thing you built (a repo, video, or write-up) and a few sentences on the hardest bug in it to jenny.hu@aimrobots.ai.